

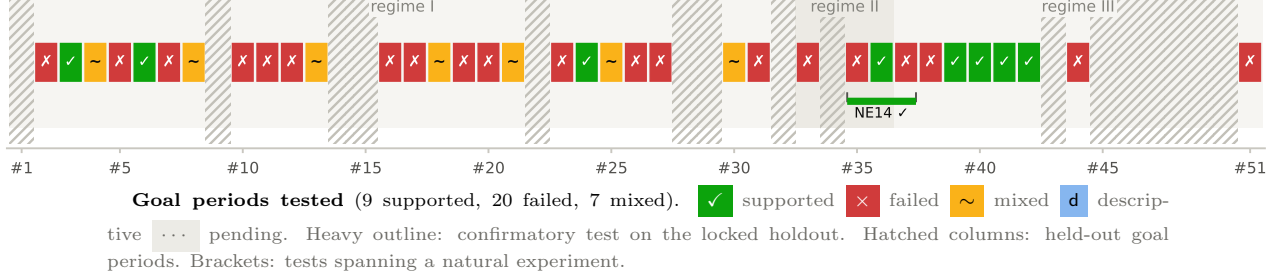
# H38: Joint silences are platform stalls

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

**Bottom line.** Joint silences are mostly chance, not stalls, but in the always-on regime two thirds of the apparent collective co-activation is agents starting and stopping together, so trim to the common running window before measuring synchrony.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

Built a shared outage table (every minute with  $\leq 1$  active agent, 389 days, causes from logs) and asked, in 35 non-holdout periods, how much of H02/H19's collective co-activation is infrastructure. **Joint silences are mostly not platform stalls:** their rate is what independent agents produce (median excess 0.008 of minutes), infrastructure-error bursts make them *less* likely (log OR  $-0.43$ ), and scaffold states sit in them near chance; village-off gaps are 84% operator-scheduled. **But in always-on regime III about two thirds of the co-activation is infrastructure** (median  $f = 0.68$ , 8 periods), mostly agents starting and stopping together; the regime-III rise in co-activation vanishes (NE14:  $+0.15 \rightarrow +0.01$ ). In regime I,  $f = 0.11$ ; talk co-activation survives everywhere. Pre-registered stall-filter test P4 failed (median  $f = 0.25$ ); synthetic data showed the agent-state estimator is the sound one. Holdout script frozen, not run.

## PHYSICAL MODEL

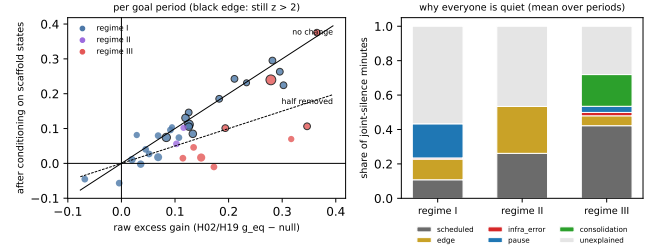
Models 01 (Curie-Weiss) and 09 (Hawkes).  $s_i(t) = \pm 1$ : agent  $i$  active in minute  $t$ ;  $h_i(b)$ : per-agent field per 30-min block;  $z(t)$ : stall indicator (operator pause, not yet started or finished, consolidating, infrastructure error), a field that switches agents off;  $\pi$ : stall fraction;  $p$ : on-rate. Stall-adjusted gain: drop off-schedule minutes, set scaffold-unavailable agent-minutes to the agent's block mean, compare with block-shift surrogates treated the same way.

$$P(\mathbf{s}) \propto \exp \left[ \frac{\beta J_0}{2N} M^2 + \sum_i (h_i(b) + \Delta_i z(t)) s_i \right], \quad M = \sum_i s_i$$

$$\text{VR} = \frac{\sum_b n_b \text{Var}_b M}{\sum_b n_b \sum_i \text{Var}_b s_i}, \quad g = 1 - \text{VR}^{-1}$$

$$\text{stalls alone: } \text{VR} - 1 \approx \frac{(N-1)\pi p}{1 - (1-\pi)p} \quad (N=12, p=0.5, \pi=0.05 \Rightarrow g \approx 0.34)$$

$$f = 1 - \frac{g_{\text{adj}} - \langle g_{\text{adj}} \rangle_{\text{null}}}{g_{\text{raw}} - \langle g_{\text{raw}} \rangle_{\text{null}}}$$



**Observables.** (a) Excess equal-time gain per goal period before and after conditioning on scaffold states (dashed: half removed). Regime-III points (red) drop to or below the dashed line, except #44 and #51; regime I (blue) stays on the identity line. Black edge: still  $z > 2$ . (b) Primary causes of joint-silence minutes, mean over periods: in regime I most have no logged reason; in regime III, operator schedule, day edges and consolidation dominate. Infrastructure errors are under 1%.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A useful negative plus one practical rule. If you run an always-on swarm with a daily start and stop, any synchrony number (mean-field coupling, market-mode eigenvalue, co-activation) is mostly your own schedule: agents boot and stop together. Trim each day to the window in which all agents run, and condition on logged consolidation and error gaps, before reading such a number as coordination; here that removes about two thirds of it. Beyond that it's mostly book-keeping: joint silences aren't a stall signal worth alerting on, error logs don't predict them, and real provider outages leave no trace here. Message-driven swarms (regime I) get nothing from this. The #44 and #51 residuals are the coupling worth studying.

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 fra/summaries/build\_summaries.py · card hypotheses/H38-platform-  
 stalls/README.md · ratings: H38 agent (suggested)

## RESULTS VS PREDICTIONS

| Prediction                                                                                                                                        | Observed                                                                  | Verdict      |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------|
| P1 JS share 0.05–0.30, median 0.10; excess only above independence; 0.008; 9 periods village-off in $\leq 6$ periods                              |                                                                           | $\times$     |
| P2 scaffold reasons in explained 0.51 (strict 0.16), > JS above chance; timer surrogate q95 in 8/35; pauses pauses lead in regime III lead in 0/9 |                                                                           | $\times$     |
| P2d village-off minutes $\geq 84\%$ 80% scheduled                                                                                                 |                                                                           | $\checkmark$ |
| P3 infra-error bursts pre-ceed JS (OR > 1)                                                                                                        | reversed: log OR $-0.43$ ; 9/30 positive, 0 significant                   | $\times$     |
| P4 stall filter removes $\geq$ half of excess gain                                                                                                | median $f = 0.25$ (III 0.63, I 0.16)                                      | $\times$     |
| Amend. 2: scaffold conditioning                                                                                                                   | $f = 0.68$ (III), 0.11 (I); H02 III chunks 13 $\rightarrow$ 9 significant | $\checkmark$ |
| P5 $\lambda_1$ drop tracks stall share ( $\rho \geq 0.6$ )                                                                                        | $\rho = 0.89$ ; drop 0.48 of lull filter's                                | partial      |
| P6 regime III–I contrast shrinks $\geq$ half                                                                                                      | $+0.125 \rightarrow +0.017$ (86%)                                         | $\checkmark$ |
| P8 agreement with Hawkes fast $n_x$ ( $\rho > 0.3$ )                                                                                              | $-0.38 \rightarrow +0.05$                                                 | $\times$     |
| P9 NE14 jump carried by stalls                                                                                                                    | $+0.15$ [0.07, 0.23] $\rightarrow +0.01$ (stall), $-0.06$ (scaffold)      | $\checkmark$ |

## GOAL PERIODS IN DETAIL

G02  $\times$  failed  
G03  $\checkmark$  supported  
G04  $\sim$  mixed  
G05  $\times$  failed  
G06  $\checkmark$  supported  
G07  $\times$  failed  
G08  $\sim$  mixed  
G10  $\times$  failed  
G11  $\times$  failed  
G12  $\times$  failed  
G13  $\sim$  mixed  
G16  $\times$  failed  
G17  $\times$  failed  
G18  $\sim$  mixed  
G19  $\times$  failed  
G20  $\times$  failed

+20 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

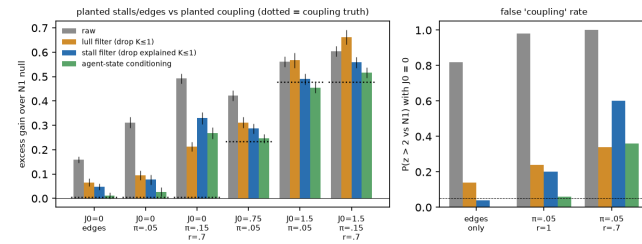
LLM API failures leave no log line. They hide among the unexplained joint silences (31% of JS minutes), so provider outages remain untested. The headline estimator (scaffold conditioning) was added after synthetic smoke runs and before any real-data gain analysis; the pre-registered stall filter failed. The surrogate test of “explained above chance” is weak for schedule-aligned stalls (G37 and G38 fail it despite  $f \approx 0.8$ ). Whether pauses count as infrastructure is a choice: regime-I waits are message-triggered, a coupling channel. Results depend on the half-of-silent rule (strict rule: explained share 0.16). The holdout has not been run.

## ROUND 2

Per-provider silence test (all agents of one lab silent while others act) for API outages; run `confirm.py` (C1–C6, including the NE21 hours reversal, where 8-h days should dilute edge-induced co-activation); study the #44 and #51 residual coupling.

*No round-2 redirects yet.*

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**Second figure.** Synthetic swarms ( $N = 12$ , 5 days; 50 replicates per condition). Stalls alone make the raw gain significant in 98% of replicates. The pre-registered stall filter leaves 20% false “coupling” at full marker recall; agent-state conditioning leaves 6% and recovers planted coupling within  $-5$  to  $+23\%$ . Both degrade when 30% of stalls are unmarked (60% and 36%).